From 500 to a General Formula: Exploring a Floor Identity

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1 THE PROBLEM AND A CHANGE OF PERSPECTIVE

The original problem. Prove that there exists a real number w such that

$$\boxed{\sum_{k=0}^{500} \left\lfloor \frac{x+2^k}{2^{k+1}} \right\rfloor = \lfloor x \rfloor - \lfloor wx \rfloor \qquad \text{for every } x \in \mathbb{R}.} \tag{1}$$

Before trying to solve the problem, let us test a special value of x. Taking $x = 2^{501}$, we have

$$\left\lfloor \frac{2^{501}+2^k}{2^{k+1}} \right\rfloor = 2^{500-k}, \qquad 0 \leqslant k \leqslant 500.$$

Therefore,

$$\sum_{k=0}^{500} \left\lfloor \frac{2^{501}+2^k}{2^{k+1}} \right\rfloor = \sum_{k=0}^{500} 2^{500-k} = 2^{501} - 1.$$

If the identity holds for every real x, we must have

$$2^{501} - 1 = 2^{501} - \left\lfloor 2^{501} w \right\rfloor,$$

and hence

$$\left\lfloor 2^{501} w \right\rfloor = 1.$$

By the definition of the floor function,

$$1 \leqslant 2^{501} w < 2.$$

Dividing by the positive number 2^{501} gives

$$\boxed{2^{-501} \leqslant w < 2^{-500}.}$$

This gives an interval for w, but can we narrow it down? Let us generalize our choice of x by taking

$$x = 2^{500+s}, \qquad s \in \mathbb{Z}_{>0}.$$

For every $0 \leqslant k \leqslant 500$, we have

$$\left\lfloor \frac{2^{500+s}+2^k}{2^{k+1}} \right\rfloor = 2^{499+s-k}.$$

Consequently,

$$\begin{aligned}\sum_{k=0}^{500}\left\lfloor\frac{2^{500+s}+2^k}{2^{k+1}}\right\rfloor &= \sum_{k=0}^{500}2^{499+s-k}\\ &= 2^{500+s}-2^{s-1}.\end{aligned}$$

If the original identity holds for every real x, then

$$\left\lfloor 2^{500+s}w\right\rfloor = 2^{s-1}.$$

By the definition of the floor function,

$$2^{s-1}\leqslant 2^{500+s}w < 2^{s-1}+1.$$

Dividing by the positive number 2^{500+s}, we obtain

$$\boxed{2^{-501}\leqslant w < 2^{-501}+2^{-(500+s)}.}$$

For $s=1$, this is precisely the interval obtained from $x=2^{501}$. As s increases, the upper bound approaches the lower bound 2^{-501}. Since the inequalities hold for every positive integer s, we conclude that

$$\boxed{w=2^{-501}.}$$

We have found the only possible value of w, but we have not yet proved that it works for every real x. Our choices of special values helped us determine w; they gave us no clear method for proving the identity for an arbitrary real x. How can we proceed?

There seems to be nothing mathematically special about 500: the summands follow the same pattern regardless of where we stop. Perhaps making the problem more general will help us find a proof. Let us therefore replace 500 by a nonnegative integer m. Since the unknown number may depend on m, we denote it by w_m.

We are thus led to a more general problem: for each $m\in\mathbb{Z}_{\geqslant 0}$, find a real number w_m such that

$$\sum_{k=0}^{m}\left\lfloor\frac{x+2^k}{2^{k+1}}\right\rfloor = \lfloor x\rfloor - \lfloor w_m x\rfloor \qquad \text{for every } x\in\mathbb{R}. \tag{2}$$

Our purpose in introducing m is not merely to find another unknown coefficient: we already know the necessary value when $m=500$. We want to discover *how to prove* the identity. At first sight, the generalized problem may seem more difficult, but it allows us to begin with $m=0$ and $m=1$. We will still choose special values of x to find candidate coefficients. The more important question is how to verify the equalities for every real x, and whether those proofs reveal a general structure.

2 EXPLORING THE GENERALIZED PROBLEM

2.1 The case $m=0$

Equation (2) becomes

$$\left\lfloor\frac{x+1}{2}\right\rfloor = \lfloor x\rfloor - \lfloor w_0 x\rfloor. \tag{3}$$

Step 1: Find a candidate.

We use the same strategy as in the original problem: choose a family of special values of x, obtain inequalities, and let the parameter tend to infinity. This time, take $x = 2n-1$ with $n \in \mathbb{Z}$. We choose $n \geqslant 1$, so that $2n-1 > 0$. Then

$$\left\lfloor \frac{(2n-1)+1}{2} \right\rfloor = n, \qquad \lfloor 2n-1 \rfloor = 2n-1.$$

If (3) holds for all x, it follows that

$$n = 2n-1-\lfloor w_0(2n-1) \rfloor,$$

and hence

$$\lfloor w_0(2n-1) \rfloor = n-1.$$

By the definition of the floor function,

$$n-1 \leqslant w_0(2n-1) < n.$$

Our choice $n \geqslant 1$ ensures that $2n-1 > 0$. Dividing by this positive number gives

$$\frac{n-1}{2n-1} \leqslant w_0 < \frac{n}{2n-1}.$$

Since the assumed identity holds for every real x, these inequalities hold for every positive integer n. Both bounds tend to $\frac{1}{2}$ as $n \to \infty$. Therefore,

$$\boxed{w_0 = \frac{1}{2}.}$$

This is a *necessary condition*. So far, we have not proved that this value works for every real x.

Step 2: Verify the candidate.

We recall the following special case of *Hermite's floor identity*:

$$\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor, \qquad y \in \mathbb{R}. \tag{4}$$

A proof is given in the appendix.

Taking $y = x/2$, we obtain

$$\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor.$$

Hence (3) holds for every real x with $w_0 = \frac{1}{2}$.

2.2 The case $m = 1$

Now the desired identity becomes

$$\left\lfloor \frac{x+1}{2} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor = \lfloor x \rfloor - \lfloor w_1 x \rfloor. \tag{5}$$

We follow the same two-step strategy.

Step 1: Find a candidate.

As in the first case, take $x = 2n-1$ with $n \in \mathbb{Z}$.

To simplify the second floor term as well, choose $n = 2t$ with $t \in \mathbb{Z}_{>0}$. Thus our test values are

$$x = 4t - 1 > 0.$$

Then

$$\left\lfloor \frac{x+1}{2} \right\rfloor = 2t, \qquad \left\lfloor \frac{x+2}{4} \right\rfloor = \left\lfloor t + \frac{1}{4} \right\rfloor = t.$$

Thus (5) implies

$$3t = 4t - 1 - \lfloor (4t-1)w_1 \rfloor,$$

or equivalently,

$$\lfloor (4t-1)w_1 \rfloor = t - 1.$$

By the definition of the floor function,

$$t - 1 \leqslant (4t-1)w_1 < t.$$

Our choice $t \geqslant 1$ ensures $4t - 1 > 0$. Dividing by this positive number gives

$$\frac{t-1}{4t-1} \leqslant w_1 < \frac{t}{4t-1}.$$

Since the assumed identity holds for every real x, these inequalities hold for every $t \in \mathbb{Z}_{>0}$.

Letting $t \to \infty$, we obtain the necessary value

$$\boxed{w_1 = \frac{1}{4}.}$$

Step 2: Verify the candidate.

Apply (4) twice. With $y = x/4$,

$$\left\lfloor \frac{x}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor = \left\lfloor \frac{x}{2} \right\rfloor. \tag{6}$$

With $y = x/2$,

$$\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor. \tag{7}$$

Adding these equalities and cancelling the common term $\lfloor x/2 \rfloor$ gives

$$\left\lfloor \frac{x}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor.$$

This proves (5) for every $x \in \mathbb{R}$ with $w_1 = \frac{1}{4}$.

What have we discovered?

The first two cases give

$$w_0 = \frac{1}{2}, \qquad w_1 = \frac{1}{4},$$

suggesting that

$$w_m = \frac{1}{2^{m+1}}$$

might work in general.

More importantly, the proof for $m = 1$ reveals a structure: applying Hermite's identity at successive scales allows intermediate terms to cancel. This suggests a way to prove the conjecture for arbitrary m.

2.3 The general case

We now prove that the proposed formula works for every nonnegative integer m.

Theorem 2.1. *For every $m \in \mathbb{Z}_{\geqslant 0}$ and every $x \in \mathbb{R}$,*

$$\boxed{\sum_{k=0}^{m} \left\lfloor \frac{x+2^k}{2^{k+1}} \right\rfloor + \left\lfloor \frac{x}{2^{m+1}} \right\rfloor = \lfloor x \rfloor .} \tag{8}$$

In particular, $w_m = 2^{-(m+1)}$ satisfies (2).

Proof. The key is again Hermite's identity

$$\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor .$$

For each $j = 1, \ldots, m+1$, take $y = x/2^j$. Then

$$\left\lfloor \frac{x}{2^j} \right\rfloor + \left\lfloor \frac{x+2^{j-1}}{2^j} \right\rfloor = \left\lfloor \frac{x}{2^{j-1}} \right\rfloor .$$

When $m = 0$, this is just one application of Hermite's identity. For larger m, the successive equalities have the following form:

$$\begin{aligned}
\left\lfloor \frac{x}{2^{m+1}} \right\rfloor + \left\lfloor \frac{x+2^m}{2^{m+1}} \right\rfloor &= \left\lfloor \frac{x}{2^m} \right\rfloor , \\
\left\lfloor \frac{x}{2^m} \right\rfloor + \left\lfloor \frac{x+2^{m-1}}{2^m} \right\rfloor &= \left\lfloor \frac{x}{2^{m-1}} \right\rfloor , \\
&\vdots \\
\left\lfloor \frac{x}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor &= \left\lfloor \frac{x}{2} \right\rfloor , \\
\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor &= \lfloor x \rfloor .
\end{aligned}$$

Add all these equations. Every intermediate term $\left\lfloor x/2^j \right\rfloor$, $1 \leqslant j \leqslant m$, appears once on each side and cancels. What remains is

$$\left\lfloor \frac{x}{2^{m+1}} \right\rfloor + \left\lfloor \frac{x+2^m}{2^{m+1}} \right\rfloor + \left\lfloor \frac{x+2^{m-1}}{2^m} \right\rfloor + \cdots + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor ,$$

which is exactly (8). □

Returning to the original problem, take $m = 500$. The identity holds for every $x \in \mathbb{R}$ with

$$\boxed{w = \frac{1}{2^{501}}.}$$

Together with the necessary condition established in Section 1, this shows that $w = 2^{-501}$ is the *unique* real number satisfying (1).

3 REFLECTION: FROM A SPECIFIC PROBLEM TO A GENERAL PROOF

Our exploration involved two different generalizations. First, with the upper limit fixed at 500, we replaced the single test value $x = 2^{501}$ by the family $x = 2^{500+s}$. The resulting intervals narrowed as s increased. Taking a limit determined the only possible value of w, but did not tell us how to prove the identity for arbitrary real x.

This difficulty motivated our second generalization: we replaced 500 by a variable m. The aim was no longer simply to find the coefficient, but to discover a method of proof. The general problem allowed us to examine $m = 0$ and $m = 1$. In each case we again chose a family of special values of x and used a limit to find a candidate. The method for finding necessary conditions was familiar; what was new was the proof of sufficiency using Hermite's identity.

The case $m = 1$ was particularly revealing. Its proof showed that applying Hermite's identity at successive scales allows intermediate terms to cancel. This provided a structure that could be repeated for arbitrary m.

Testing special values can help us discover what a coefficient must be, but does not establish an identity for every real number. The first generalization helped us determine the necessary value; the second helped us understand why the proposed value works.

Sometimes, making a problem more general is precisely what makes it easier to prove.

A A SPECIAL CASE OF HERMITE'S FLOOR IDENTITY

We prove that, for every $y \in \mathbb{R}$,

$$\boxed{\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor .}$$

Proof. Let $n = \lfloor y \rfloor$. Then $y = n + r$ for some $0 \leqslant r < 1$. Hence

$$\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = 2n + \left\lfloor r + \frac{1}{2} \right\rfloor , \qquad \lfloor 2y \rfloor = 2n + \lfloor 2r \rfloor .$$

It remains to show that $\left\lfloor r + \frac{1}{2} \right\rfloor = \lfloor 2r \rfloor$.

If $0 \leqslant r < \frac{1}{2}$, both sides equal 0. If $\frac{1}{2} \leqslant r < 1$, both sides equal 1. Thus the identity holds for every $y \in \mathbb{R}$. □